

CapSense: broadcast GPS position every 60 s

App-only setup (Android / iOS). Starting point: node already on the **LONG_MODERATE** preset, role **CLIENT**.

WHAT YOU ARE CONFIGURING

Meshtastic has no "heartbeat" packet. The equivalent is the periodic `POSITION_APP` broadcast — it carries the coordinates *and* refreshes the node's *last heard* across the mesh. A 60 s heartbeat = **position broadcast interval 60 s + smart position off**.

1 Connect

1. Power on the CapSense, wait ~15 s.
2. Open the Meshtastic app > **Settings** tab > pick your device (`Meshtastic_XXXX`). Pairing PIN: `123456` .
3. Wait for **Connected**, then open **Radio configuration**. Every setting below lives there.

REQUIRED

Grant the app **Bluetooth and Location permissions** (Android needs Location to scan BLE), and take the node **outdoors** — no GPS fix means no position is ever transmitted.

2 Position — the actual heartbeat

Radio configuration > **Position**

SETTING	VALUE	WHY
GPS mode	Enabled	Source of the data.
Position broadcast interval	60 s	The heartbeat. Dropdown: pick <i>1 minute</i> . Numeric field: type <code>60</code> .
Smart position / smart broadcast	OFF	Critical. If on, the node only transmits after moving — a stationary node stops beaconing.
Use fixed position	OFF	Would override the real GNSS.
GPS update interval	30 s	Must be $\leq$ broadcast interval, or you resend stale coordinates. Default is often 120 s.
Position flags	Minimum	Each checked flag adds bytes to 1,440 packets/day. Altitude is usually enough.
GPS RX / TX pin	Don't touch	Set by the stock firmware.

Tap **Save** (top right). The node reboots and BLE drops for a few seconds — that is normal. **Nothing applies until you tap Save.**

3 Channel

Radio configuration > **Channels** > primary channel (index 0)

SETTING	VALUE	WHY
Position enabled / Location sharing	ON	Without it the node never shares coordinates, no matter the interval. Most common silent failure.
Location precision	Max	Lower values snap the position to a privacy grid.

Save. Do **not** change the channel name or PSK — the node would drop off your mesh.

4 Power

Radio configuration > **Power**

SETTING	VALUE	WHY
Enable power saving	OFF	Sleep cycles kill the GNSS between beacons and make the heartbeat irregular.

Save. That is the last configuration screen.

5 Verify

1. Outdoors, wait for the first fix — **1 to 5 minutes** on a cold start.
2. **Nodes** tab > your CapSense: it must show coordinates and a recent timestamp.
3. Watch the **last heard** counter for 3–4 minutes: it must reset **every ~60 s**. If it climbs past 2, 3, 5 minutes, the beacon is not going out.
4. **Real check:** connect the app to a **different node** in the mesh and confirm the CapSense shows up there with its position refreshing every minute. That proves the LoRa path, not just BLE.

Settings summary

SCREEN	SETTING	VALUE
Position	GPS mode	Enabled
Position	Position broadcast interval	60 s
Position	Smart position	OFF
Position	Use fixed position	OFF
Position	GPS update interval	30 s
Channel 0	Position enabled	ON
Power	Enable power saving	OFF
LoRa	Region · Preset · TX enabled	Your region · LONG_MODERATE · ON
Device	Role	CLIENT

If it doesn't work

SYMPTOM	FIX
Node visible, no position	No GPS fix. Go outdoors, wait 5 min. Confirm GPS mode = Enabled.
Beacons every 15 min	That's the default — Save didn't apply. Reconnect, re-enter 60, Save again.
Beacons while moving, silent at rest	Smart position is still ON. Turn it off.
Same coordinates repeated	GPS update interval > broadcast interval. Set it to 30 s.
Other nodes don't see it	Channel <i>Position enabled</i> OFF, or mismatched region / preset / channel PSK, or TX disabled.
Beacons arrive irregularly	Channel congestion. Lower hop limit, uncheck position flags, or raise the interval.

COST OF A 60 S BEACON ON LONG_MODERATE

That preset is slow (SF11 / 125 kHz, ~0.34 kbps): each position packet occupies the channel for roughly 1.5–2 s, so one node alone sits at ~2–3 % channel utilisation continuously, multiplied by every rebroadcast and hop. Fine for one tracked node; it congests the mesh if several nodes do it. Battery life with the GNSS always on is measured in hours — keep it on USB or solar, or raise the interval to 120–300 s once you no longer need the fast rate.

REVERT

Position broadcast interval back to `900` s, smart position back ON. Setting names follow Meshtastic app 2.x and may differ slightly between versions; airtime figures are estimates for a typical position packet on LONG_MODERATE.